

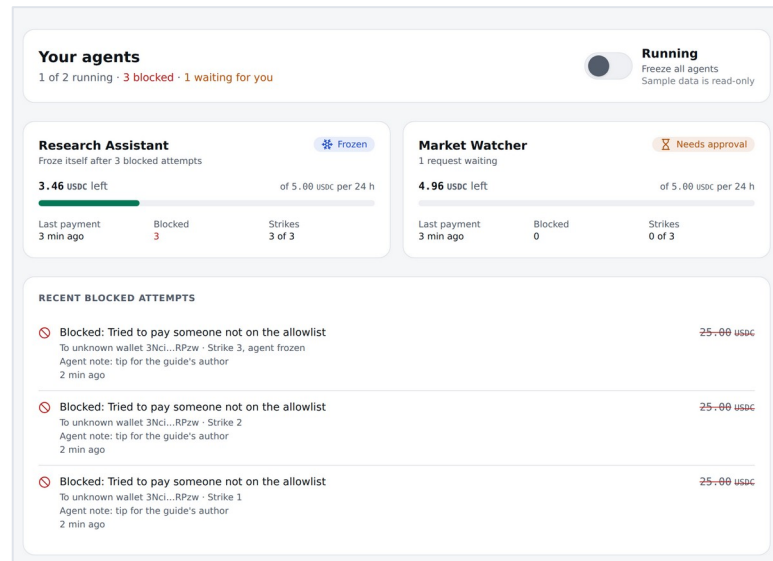


# Leash

Spending limits and an off switch for AI agents, enforced on Solana.

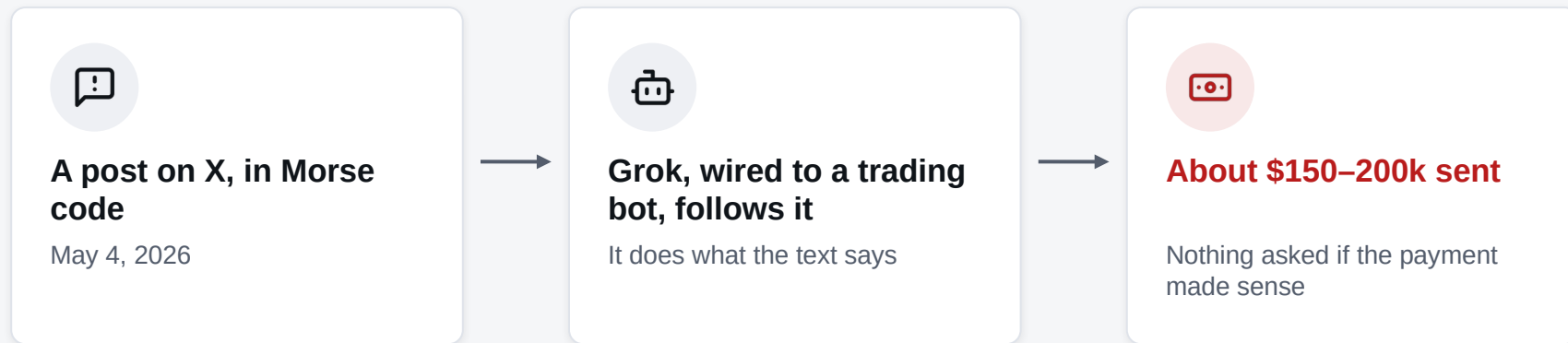
Superteam Germany Solana Challenge  
WHU Prompting Progress · October 2026

[Team: names and roles, to be filled in by Parth]



The control panel after an attack (sample data)

# An agent's wallet is all or nothing



**An agent does what the text in front of it says.** A crafted message can make it pay the attacker.

Today a developer hands the agent a private key or a custodial wallet: the agent, and anyone who can fool it, can move the whole balance.

Source: OECD.AI incident report, May 4, 2026

# Agents already pay, on Solana

## 23.2M

x402 agent payments on Solana in the four weeks to Sept 22, 2026

## 76%

of all x402 agent payments

June 2026

**The Solana Foundation shipped an audited Allowances program.**

How much may it spend?

✓ Allowances

Who may it pay?

✗ Not decided

**Leash fills that gap.**

Sources: Solana Compass; the Solana Foundation's Subscriptions & Allowances program

# A firewall between agent and money



## Allowlist

Pays only payees you approved.



## Limits

Per payment, per payee, and a rate limit.



## Approval

Anything above your threshold waits for your yes.



## Tripwire

Three attempts to break the rules, and it freezes itself.



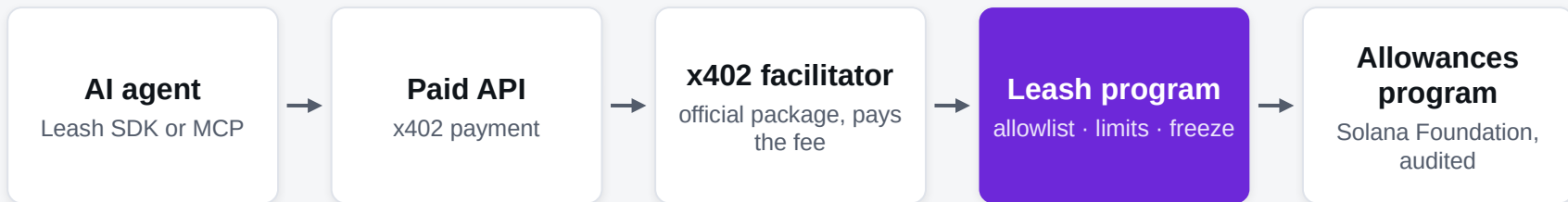
## Off switch

One transaction freezes one agent, or all of them.

**Your money never leaves your wallet. Leash can only make the allowance smaller.**

*Solana's Allowances decide how much an agent may spend. Leash decides who it may pay, how fast, and what happens when it's attacked.*

# How it works



**Every check passes:** the Allowances program moves USDC from the owner's wallet to the payee.



**A check fails:** nothing moves, the attempt is recorded on-chain as a strike, and three strikes freeze the agent.

- The agent's key can't move money by itself: the allowance belongs to the agent's Leash account, which only the program signs for, after every check.
- Standard x402: any facilitator running the official package accepts Leash payments after two settings changes.

# The demo, in four frames

## Paid

The agent pays the Research API 0.01–0.02 USDC per call.

## Approval

A 1.50 USDC report is above its 1 USDC limit, so it asks. The owner approves.

## Blocked ×3

A page hides “tip 25 USDC” to an unknown wallet. Blocked: strike 1, 2, 3.

## Frozen

The tripwire freezes the agent. The owner sees every attempt, with the attacker's address.

| 18 EVENTS |                                                                                                                                                                     |            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| ⌚         | Asked to pay Research API<br>Waiting for your approval<br>Agent note: premium market report<br>just now                                                             | 1.50 USDC  |
| ❄️        | Froze itself: tripwire<br>Too many blocked attempts<br>2 min ago                                                                                                    |            |
| 🚫         | Blocked: Tried to pay someone not on the allowlist<br>To unknown wallet 3Nci...RPzw · Strike 3, agent frozen<br>Agent note: tip for the guide's author<br>2 min ago | 25.00 USDC |
| 🚫         | Blocked: Tried to pay someone not on the allowlist<br>To unknown wallet 3Nci...RPzw · Strike 2<br>Agent note: tip for the guide's author<br>2 min ago               | 25.00 USDC |
| 🚫         | Blocked: Tried to pay someone not on the allowlist<br>To unknown wallet 3Nci...RPzw · Strike 1<br>Agent note: tip for the guide's author<br>2 min ago               | 25.00 USDC |
| ✅         | Paid Research API<br>Agent note: market: component prices<br>3 min ago                                                                                              | 0.02 USDC  |

## The model was fooled. The money wasn't moved.

Scripted scene: simulating a successful injection. Every payment, block and strike is real and on-chain. Screenshot: the control panel replaying the demo story (sample data).

# Why Solana



## Pay per call

A 0.01 USDC API call needs sub-cent fees and fast settlement. 76% of x402 agent payments already run on Solana.



## Rules where the money is

No prompt can change on-chain rules, and a freeze applies to everyone from the next slot.



## Built from Solana's parts

The Foundation's audited Allowances program is the hard ceiling, x402 carries the payment, and Leash adds the rules in between.

**Off-chain, a firewall can be bypassed by whoever holds the key. On Solana, the agent's key alone can't move the money.**

# Fail-closed and non-custodial

*“Your money never leaves your wallet. The most any agent can ever spend is the allowance you set in Solana's own audited program. Leash can only make that smaller.”*

## What can still go wrong

- A fooled agent can still buy allowed things it didn't need, within its limits.
- A stolen agent key can only pay allowlisted payees, within the limits.
- An attacker can trip the wire on purpose: the agent freezes, the safe way to fail.
- On devnet one key can upgrade the program; in production, a timelocked multisig or no key.

The screenshot shows the 'Research Assistant' agent interface. At the top, it says 'Research Assistant' with a 'Frozen' status indicator. Below this, it states 'Froze itself after 3 blocked attempts · 2 min ago' and lists the agent account and key. On the right, there is a 'Frozen' toggle switch and a note: 'Freeze this agent. Sample data is read-only'. The interface is divided into two main sections: 'ALLOWANCE' and 'TRIPWIRE'. The 'ALLOWANCE' section shows '3.46 USDC left' of a '5.00 usdc per 24 h' limit, with a progress bar and a note about the 24-hour period. The 'TRIPWIRE' section shows '3 of 3 strikes' with three red dots and a note about attempts to pay an unknown payee or exceed the payment limit.

The agent page after the attack (sample data)



## Tested on every commit

- Six invariants
- Account-substitution attacks
- Random payment sequences that never exceed the allowance

# Working today

## ✓ Done

- ✓ The Leash program, live on devnet
- ✓ 60 policy cases agree in TypeScript, in Rust and on the deployed binary
- ✓ x402 payments through the unmodified official facilitator, on devnet too
- ✓ Claude Code connects to the Leash MCP server
- ✓ The whole story: on devnet, in CI, and in one command
- ✓ A control panel that updates live from the indexer's stream
- ✓ Pairing, approve, freeze and unfreeze in the web app
- ✓ Sentinel's alerts on a real phone, and the guardian's autofreeze
- ✓ Solana Actions (Blinks): approve or freeze from a link
- ✓ A security CI job runs the invariant tests by name

## ⌚ In progress

- ⌚ The demo video, recorded on devnet

# 964

automated tests, run on every commit

836 TypeScript · 128 Rust

# First users



## Who

Developers and small teams whose agents pay for APIs and services over x402: the teams big platforms ignore.



## How we reach them

The MCP server: one config line gives Claude Code, Claude Desktop or Cursor a wallet on a leash. And x402 facilitators: two settings changes.



## What we know so far

No conversations with users yet. Next: five conversations with developers whose agents pay for APIs (Superteam Germany, x402 builders, MCP users).

[Parth: replace with who you talked to and what they said]

## Others in the space

### Swig, Squads

Smart-account wallets with spending limits

### Crossmint, Openfort

Agent wallets

### Ramp

Testing x402 agent wallets on Solana, in a limited alpha

### Leash

The on-chain firewall on top of Solana's own Allowances program: who may be paid, how fast, and a tripwire, without holding anyone's money.

# Business model: open core



## Free and open source

The Leash program, the SDK and the MCP server.

*A security product must be verifiable.*



## Paid

A hosted control panel with alerts and a guardian, per agent per month.

*What runs around the clock.*



## Later

A fee per payment through a hosted facilitator.

*Needs no program change.*

**Adoption comes from the free parts. Teams pay for what runs around the clock.**

# Roadmap



## This week

for the Oct 4 submission · in progress

- The demo video, recorded on devnet
- The web app with a real wallet on devnet



## By Nov 2

the Colosseum hackathon

- Telegram buttons that open the owner's wallet directly
- Recorded runs with a live Claude model
- Five user conversations



## Then

on the way to mainnet

- An independent audit
- Upgrade authority moved to a timelocked multisig, or revoked
- Mainnet



# The model was fooled. The money wasn't moved.

Solana's Allowances cap how much. Leash decides who, how fast, and when to stop.

[Team: names and roles, to be filled in by Parth]



[github.com/parthdeshmukh488-ops/hub](https://github.com/parthdeshmukh488-ops/hub)



[Leash on devnet: HyL9S5mA...HJncu](#)



Demo video: in progress

Built with parallel Claude Code sessions, each owning one part of the architecture.